

Question 1: Explain the concept and need for cyber security. What are the pros and cons (benefits and disadvantages) of digital security?

Answer:

Concept and Need for Cyber Security: Information Technology is a dual-edged sword. While it is used for the betterment of mankind in areas like telecommunications, governance, and finance, it can also be used for disruptive purposes. Cyber security provides protection against the use of information technology for such disruptive purposes. It involves the technologies, processes, and practices designed to protect computers, computer networks, and computer systems from cyber-attacks.

According to Section 2(nb) of the Information Technology Act 2000, cyber security is defined as the protection of information, devices, equipment, computer, computer resource, and communication devices, as well as information stored therein, from unauthorized access, disclosure, disruption, modification, and destruction. Due to increasing cybercrimes, it is highly needed by countries to curb exploitation and protect their data.

Pros (Benefits) of Digital Security:

- It helps in protecting personal information stored in devices.
- Suspicious or unauthorized access to devices can be blocked, preventing possible harm.
- Security based on biometrics provides a higher degree of protection against attacks because biometric information is difficult to steal.
- It enables individuals to fearlessly communicate, transact, and work in an online mode.
- It protects the computer from crashing or slowing down, thereby protecting business transactions and communications happening over the network.
- It helps in fostering the economy of the State by cutting down on many costs associated with data breaches.

Cons (Disadvantages) of Digital Security:

- Availing services or procuring tools for digital security can be a costly affair.
- Web services or security tools may or may not be compatible with the specific device of the user.
- Digital security services or tools can sometimes be difficult to configure and need to be updated regularly.
- These services or tools may slow down the functioning of the user's device or occasionally intervene with the normal functioning of another program.

Question 2: Discuss the various security issues and threats emerging in Cyberspace (e.g., Distributed Denial of Service (DDoS) attack, Malwares, Crypto-jacking, Cyber-physical attack, Phishing).

Answer:

Threats to cyber security are constantly evolving. The most common security issues and threats witnessed in cyberspace include:

1. **Unauthorised access:** This means accessing a computer, network, system, or device without permission from the authorized owner. For example, stealing someone's pen drive to access data or guessing email passwords to read private emails.
2. **Distributed Denial of Service (DDoS) Attack:** This attack aims to adversely affect the functioning of a website by sending multiple requests that are beyond its control to handle. It floods a machine or network resource with unnecessary traffic to make it unavailable to intended users.
3. **Malwares:** These are malicious software designed to harm a network or device. Types of malware include:
 - *Botnets:* Collections of infected devices connected through the internet, controlled to carry out cyber-attacks.
 - *Ransomwares:* Malware that encrypts the victim's files or data and demands a ransom to decrypt or permit access to them.
 - *Trojan:* A malicious software that prima facie looks legitimate but is intended to steal, harm, or damage the victim's device.
 - *Virus:* A self-replicating malicious program capable of spreading across networks and affecting the functioning of a computer.
 - *Spywares:* Software installed secretly to spy on and gather information about the victim.
4. **Social Engineering attacks:** These attacks rely on the psychological manipulation of victims, tricking them into revealing sensitive information or clicking on malicious links.

5. **Phishing:** An activity where the victim is deceived into revealing sensitive information (like usernames, passwords, or credit card details) by an attacker disguising themselves as a trustworthy entity (e.g., a fake email from a company's Finance Officer).
6. **Crypto-jacking:** The unauthorized use of a victim's device to secretly mine cryptocurrency. This takes a heavy toll on the device's CPU and affects its functioning.
7. **Exploiting vulnerability:** Attackers exploit a flaw or weakness in the security measures of a device to gain access or cause harm.
8. **Cyber physical attacks:** These are cyber-attacks that breach security and directly impact the physical environment. Examples include shutting down cyber-controlled cameras/lights, taking over water pumps, or manipulating cooling systems in nuclear reactors.
9. **Internet of Things (IoT) attacks:** Embedded devices connected to a network (like fax machines) are exploited due to vulnerabilities and can be hacked to steal data.
10. **Web Jacking:** Inspired by the term hijacking, it means taking over the access and control of a website to trick visitors or deface the site.
11. **Drive by download:** When a victim browses a compromised legitimate website, it secretly installs a malicious payload (like ransomware) into the victim's device.

Question 3: What are the different challenges to cyber security? How can technology be used to provide solutions to cyber-attacks (like unauthorized access, malwares, etc.)?

Answer:

Different Challenges to Cyber Security: Challenges to cyber security exist in the form of unauthorized access, exploiting vulnerabilities, cyber-physical attacks, IoT attacks, Web jacking, drive-by downloads, crypto-jacking, social engineering attacks, malwares (like trojans, viruses, worms, and spywares), and distributed denial of service (DDoS) attacks. These threats are constantly evolving and are harmful to an organization as they steal, harm, or corrupt information stored in systems.

Technology's Answers to Cyber-Attacks: Technology provides various solutions and measures to prevent and mitigate these cyber security challenges:

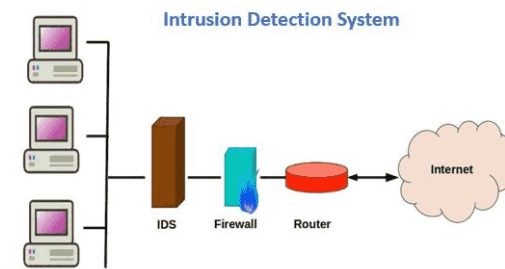
1. **Unauthorized Access:** Strong passwords, endpoint security, two-factor authentication, physical security practices, and monitoring user activity are common practices employed for protection.
2. **Distributed Denial of Service (DDoS) Attack:** Various infocom security tools are used, including anti-malware, firewalls, spam filtering, switches and routers, Intrusion Prevention Systems (IPS), DDoS defense systems, and content delivery networks.
3. **Malwares:**
 - *Botnets:* Virtual Private Networks (VPNs), secured network architecture, and traffic management tools help in detecting and preventing botnet attacks.
 - *Ransomwares:* Backing up data and disabling unnecessary ports or services (such as RDP) help prevent ransomware attacks.
 - *Trojans:* Robust firewalls, anti-spywares, and maintaining cyber hygiene are helpful.
 - *Viruses:* Malware removal software and automatic scans reduce the risk of virus attacks.
 - *Worms:* Internet connection firewalls, network intrusion detection software, Domain Message Authentication Reporting (DMARC), and Domain Name System Security Extension (DNSSEC) can be helpful.
 - *Spywares:* Anti-spywares, sandbox protection, and ad pop-up blockers play a significant role in averting spywares.
4. **Social Engineering Attacks:** Network traffic analysis, firewalls, updated software, spam protection guards, and updated blacklists aid in prevention.
5. **Phishing:** SPAM filters, web filters, patching, and the use of SSL certificates to secure traffic assist in combating phishing attacks.
6. **Crypto-jacking:** Anti-crypto mining extensions, endpoint protection, web filtering tools, and network monitoring solutions prevent crypto-jacking.
7. **Exploiting Vulnerability:** Patching operating systems, enabling SMB signing, and network segmentation (which limits access to systems) reduce the chances of exploitation.
8. **Cyber-Physical Attacks:** Adequate control on physical access, securing cyber nodes, enabling remote access only when necessary, and two-factor authentication are beneficial.
9. **IoT Attacks:** Firmware updates with cryptographic signatures, proper identity management, and hardened toolchains provide protection.

Question 4: Write short notes on: (a) Intrusion Detection System (b) Malwares.

Answer:

(a) Intrusion Detection System (IDS): Intrusion detection systems monitor network traffic and generate alerts in the case of suspicious activity that tends to harm cyber security. Some of these systems are even capable of preventing cyber threats.

- **Types:** Such systems may be network-based, host-based, hybrid, application-based, or protocol-based.
- **Methods of Detection:** They use either *signature-based* detection (which detects intrusions based on known patterns or malicious instruction sequences) or *anomaly-based* detection (which relies on a trustful activity model using machine learning to alert anything dissimilar as suspicious).
- **Benefits:** With the rise of the IoT environment, the use of IDS has grown multifold. They help ensure IT regulatory compliance, maintain security standards, measure cyber-attacks, and raise alarms against malwares like spywares, keyloggers, and unintentional leakage.
- **Flaws:** They are susceptible to raising false alarms, are often unable to read encrypted packets, need continual updates, and generally require an expert engineer to manage them.



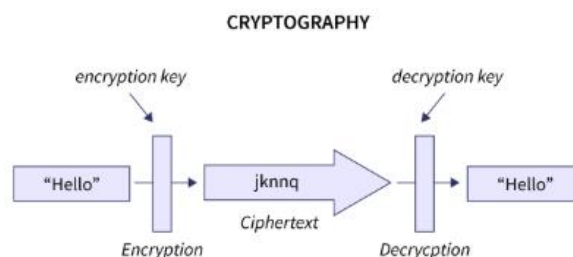
(b) Malwares: Malwares are malicious software programs that are specifically designed to harm a network or device. Under Section 43 Explanation 1 (a) and (b) of the Information Technology Act 2000, malwares are categorized as "computer contaminants" because they are capable of modifying, destroying, or transmitting data, or usurping the normal operation of a computer system. Common types of malwares include:

- **Botnets:** Collections of devices connected through the internet that are infected and controlled to carry out cyber-attacks.
- **Ransomware:** Malware that encrypts the victim's files or data and demands a ransom payment to decrypt or permit access to them.
- **Trojan:** A malicious software that prima facie looks legitimate but is intended to secretly steal, harm, or damage the victim's device.
- **Virus:** A malicious program designed to affect the functioning of a computer, capable of self-replicating and spreading across networks.
- **Worms:** Similar to viruses, these can overwrite system and personal files (e.g., the infamous ILOVEYOU worm in the year 2000).
- **Spywares:** Malicious software installed in a victim's device without their knowledge to secretly spy on and gather information.

Question 5: What is meant by Cryptography? Explain the five key functions of cryptography.

Answer:

What is Cryptography? Cryptography is the art of devising ciphers to secure computer systems by encrypting sensitive records and messages in transit and in storage. In cryptography, the message or data to be encrypted (known as the plaintext) is transformed by a function that is parameterized by a KEY. The output of this encryption process is known as the cipher text, which is then transmitted through an insecure communication channel. Mathematically, it is represented as $C = E_k(P)$, meaning the encryption of the plaintext P using key K gives the cipher text C .



The Five Key Functions of Cryptography: Base cryptographic functions provide the most flexible means of developing cryptography applications. The five key functions of cryptography are:

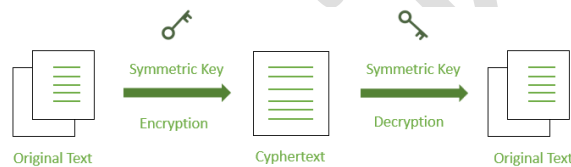
1. **Privacy/Confidentiality:** Making sure that no one can peruse (read) the message except the intended receiver.
2. **Authentication:** The process of showing and establishing one's identity.
3. **Integrity:** Assuring the receiver that the received message has not been distorted or altered in any manner from its initial form.
4. **Non-repudiation:** A process or a way to prove that the sender really sent this message, preventing them from denying it later.
5. **Key exchange:** The approach or mechanism by which crypto keys are shared securely between the sender and the receiver.

Question 6: What is Encryption? Distinguish between Symmetric key and Public key encryption schemes.

Answer:

What is Encryption? Encryption is a common method of protecting information transmitted over unreliable links. The mechanism works in the following steps:

- The information is encrypted (encoded) from its initial readable form (called clear text) into an internal, unreadable form (called cipher text).
- The cipher text can then be stored or safely transmitted over unprotected channels.
- Finally, the receiver must decrypt (decode) the cipher text back into clear text to understand its meaning.



Distinction Between Symmetric Key and Public Key Encryption Schemes:

Feature	Symmetric Key Cryptography (Private-Key)	Public Key Cryptography (Asymmetric)
Key Usage	A system where one secret key is shared and used for both encryption and decryption.	A cryptosystem where two different keys (a public key and a private key) are used for encryption and decryption.
Encryption/Decryption Process	The sender and receiver must have a pre-shared key, which is kept undisclosed from all other parties.	The sender uses the public key to carry out encryption, whereas the receiver uses the private key (kept secret) for decryption.
Key Transmission	The secret key must remain secret at both ends, making key distribution across large networks a challenge.	The secret (private) keys do not have to be transmitted or revealed to anyone, providing increased security.
Speed and Efficiency	Symmetric-key ciphers have high rates of data throughput and are very fast.	Throughput rates for public-key encryption are several orders of magnitude slower than symmetric-key schemes.
Algorithm Examples	Data Encryption Standard (DES) and Advanced Encryption Standard (AES).	RSA Algorithm.

Question 7: Discuss the advantages and disadvantages of Symmetric-key cryptography and Public-key cryptography.

Answer:

Symmetric-Key Cryptography

In Symmetric-key cryptography, a single shared key is used for both encryption and decryption.

Advantages:

- **High Speed:** Symmetric-key ciphers can be designed to have high rates of data throughput. Hardware implementations can achieve hundreds of megabytes per second, and software can attain throughput in the megabytes per second range.
- **Shorter Keys:** Keys for symmetric-key ciphers are relatively short.
- **Versatile Primitives:** They can be employed as primitives to construct various cryptographic mechanisms, including pseudorandom number generators, hash functions, and computationally efficient digital signature schemes.
- **Stronger Ciphers:** Simple transformations that are easy to analyze but weak on their own can be composed to construct strong product ciphers.
- **Extensive History:** It has an extensive history of use and research, particularly subsequent to the invention of the digital computer and the Data Encryption Standard (DES) in the early 1970s.

Disadvantages:

- **Secrecy Requirement:** In a two-party communication, the key must remain an absolute secret at both ends.
- **Key Management:** In a large network, there are many key pairs to be managed, which requires the use of an unconditionally trusted Third Party (TTP).
- **Frequent Key Changes:** Sound cryptographic practice dictates that the key must be changed frequently, perhaps for each communication session.
- **Digital Signatures Constraints:** Digital signature mechanisms arising from symmetric-key encryption typically require either large keys for verification or the use of a TTP.

Public-Key Cryptography (Asymmetric)

In Public-key cryptography, two different keys are used: a public key for encryption and a private key for decryption.

Advantages:

- **Simplified Secrecy:** Only the private key must be kept secret (though the authenticity of the public key must be guaranteed).
- **Easier Key Administration:** The administration of keys on a network requires the presence of only a *functionally* trusted TTP (which might only be required off-line), rather than an unconditionally trusted TTP.
- **Long-term Validity:** A private/public key pair may remain unchanged for considerable periods of time, even spanning several years or many sessions.
- **Efficient Digital Signatures:** Many public-key schemes yield relatively efficient digital signature mechanisms, as the key used for public verification is typically much smaller than its symmetric-key counterpart.
- **Fewer Keys:** In a large network, the total number of keys necessary is considerably smaller than in a symmetric-key scenario.

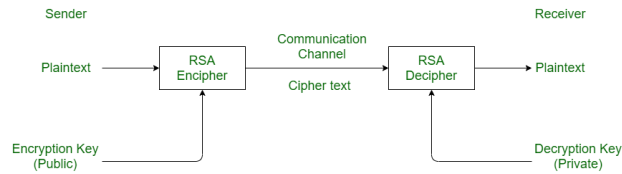
Disadvantages:

- **Slower Throughput:** Throughput rates for the most popular public-key encryption methods are several orders of magnitude slower than the best-known symmetric-key schemes.
- **Larger Key Sizes:** Key sizes are typically much larger than those required for symmetric-key encryption, and the size of public-key signatures is also larger.
- **No Absolute Proof of Security:** No public-key scheme has been completely proven to be secure; their security is primarily based on the presumed difficulty of solving a small set of complex number-theoretic problems.
- **Shorter History:** Public-key cryptography does not have as extensive a history as symmetric-key encryption, having only been discovered in the mid-1970s.

Question 8: Discuss the RSA algorithm with the help of an example.

Answer:

What is the RSA Algorithm? RSA gets its name from the initials of its three discoverers (Rivest, Shamir, and Adleman) at M.I.T.. It is a highly popular asymmetric cryptography algorithm because it uses two different keys (public and private) for encoding and decoding. The entire algorithm is based on the modular mathematics of Number theory.



Steps of the RSA Algorithm: The determination of the variables n , d , and e for key generation is prescribed in the following mathematical way:

1. **Select Primes:** Choose two large prime numbers, p and q , each typically greater than $10^{\{100\}}$.
2. **Compute n and $\phi(n)$:** * Calculate the modulus: $n = p \times q$.
 ○ Calculate the totient function: $\phi(n) = (p - 1)(q - 1)$.
3. **Choose Private Key (d):** Assume a number d to be a large, random integer that is relatively prime to $\phi(n)$.
4. **Choose Public Key (e):** Calculate e such that $e \times d \pmod{\phi(n)} = 1$. (This means e and d are multiplicative inverses modulo $\phi(n)$).
5. **Key Pair Distribution:** Since the key (e, n) is public, only the number d in the decryption pair (d, n) is kept private.

Encryption and Decryption Process: These parameters are used to encipher a plaintext P (where $0 \leq P < n$). If the plaintext is longer, it must be broken into strings smaller than n .

- **Encryption:** The Cipher text (C) is obtained using the public key equation: $C = P^e \pmod{n}$.
- **Decryption:** The Cipher text (C) may then be decrypted back to plaintext using the private key equation: $P = C^d \pmod{n}$.

These steps ensure that encryption and decryption are exact inverses of each other.

Security of RSA: The widespread use of RSA has tremendously increased because no successful breaking of the algorithm has been reported yet. A cryptanalyst would presumably have to use complex factoring to derive the private keys from n and e , which are publicly known.

Question 9: Explain the types of conventional ciphers. Specifically, explain Substitution cipher and Transposition cipher with examples.

Answer:

Conventionally, there are two main types of ciphers used in cryptography.

1. Substitution Ciphers: In a substitution cipher, another letter or a group of letters replaces each letter or group of letters in the plaintext to disguise it.

- **Example (Caesar Cipher):** One of the oldest known substitution ciphers is the Caesar Cipher, attributed to Julius Caesar. For example, using this cipher with a shift of 3, the word "attack" becomes "DWWDFN".
- **Example (One-Time Pad):** A substitution cipher can be made unbreakable by using a long, non-repeating key called a one-time pad. The key is often formed using words from a book starting from a specific place known to both the sender and receiver, and the message is encoded using XOR on the ASCII encoding of the letters.

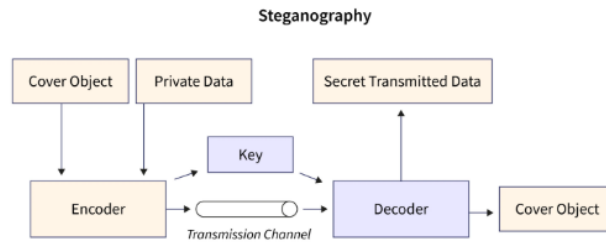
2. Transposition Ciphers: A transposition cipher operates by reordering the plaintext symbols. Whereas substitution ciphers try to disguise the symbols by replacing them, transposition ciphers preserve the original plaintext symbols but completely change their order.

- **Example (Columnar Transposition):** In this method, a keyword is used to determine the column order.
 - **Keyword:** CONSULT (Column numbers: 1 4 3 5 7 2 6).
 - **Plaintext:** "THE PEOPLE'S UNIVERSITY".
 - The plaintext is written out in rows under the columns, and the cipher text is generated by reading down the columns in the numerical order specified by the keyword.

Question 10: What is steganography? Give any two examples where the steganography technique is used.

Answer:

What is Steganography? Steganography is one of the techniques of hiding secret data within a non-secret, ordinary file so that it manages to avoid detection. In modern digital steganography, the secret data is first encrypted or obfuscated, and then it is inserted using a special algorithm into data that is part of a particular file format, such as a JPEG image, audio, or video file. Because the data is hidden out of sight and also encrypted, it becomes exceptionally challenging for an attacker to first locate the data and then decrypt it.



Examples/Uses of Steganography:

Some of the well-known uses of steganography around the world include:

1. **Historical/Physical Steganography:** In ancient times, the head of a messenger was shaved and a tattoo of the secret message was done on the head. After the hair grew back, the messenger was sent, and the recipient shaved the hair again to read the tattooed message. Other historical physical examples include the use of disappearing ink and microdots.
2. **Modern/Digital Steganography:** Osama bin Laden's pre-recorded videos that were re-played on TV stations around the world contained hidden messages embedded within them. Another example of modern steganography is hiding data in the bits that represent the same color pixels repeated in a row within an image file, creating noise patterns that are invisible to the human eye.

Question 11: Explain the concept and uses of Digital Signatures and Digital Certificates.

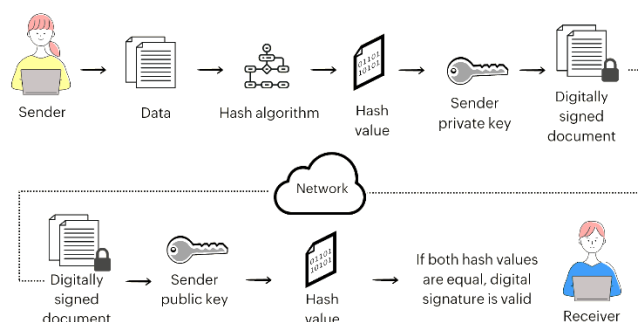
Answer:

Concept of Digital Signatures: A Digital Signature is a technique by which it is possible to secure electronic information in such a way that the originator of the information, as well as the integrity of the information, can be verified. For a computerized message system to replace the physical transport of paper and ink documents, handwritten signatures have to be replaced by Digital Signatures. It follows an approach of using a hash function, i.e., the usage of a private key and a public key, which is a two-way protection system. It is highly secure as it is encrypted and unlikely to be tampered with.

Uses of Digital Signatures:

Basically, what is needed is a system by which one party can send a "signed" message to another party in such a way that it fulfills three main conditions:

- The receiver can verify the claimed identity of the sender.
- The sender cannot repudiate the contents of the message (Non-repudiation).
- The receiver cannot possibly have concocted the message himself/herself.



Concept of Digital Certificates: Digital Certificates are provided by Certified Authorities (CAs) to authenticate that a particular site or entity is globally secured. In a digital signature system, a trusted third party (TTP), popularly called a CA, guarantees the relationship between a user's identity and their public keys.

- **Contents:** Digital certificates contain details such as the Issuer, Issued to, organization name, organization unit, validity, Version, Public Keys, Thumbprint, algorithms, etc..
- **CAs in India:** Popular Indian CAs include SafeScript Ltd, TCS, IDRBT, MTNL Ltd, and NIC.

Question 12: "Authentication is one of the most important of all information security objectives." Critically examine the statement.

Answer:

Critical Examination of Authentication: The statement holds true because authentication provides the fundamental guarantee that entities are who they claim to be, or that information has not been manipulated by unauthorized parties. Indeed, authentication is one of the most important of all information security objectives.

Separation of Secrecy and Authentication: Until the mid-1970s, it was generally believed that secrecy and authentication were intrinsically connected. However, with the discovery of hash functions and digital signatures, it was realized that secrecy and authentication are truly separate and independent information security objectives.

For example, if a two-party communication takes place between "Jack" and "Bond" residing in different countries, the host countries might not permit secrecy on the channel and might want the ability to monitor all communications. In this scenario, encryption for secrecy cannot be used. However, Jack and Bond would still like to be assured of the identity of each other, and of the integrity and origin of the information they send and receive. This proves that authentication is essential even when secrecy is not allowed.

Forms of Authentication:

- **Entity Authentication (Real-time):** If communicating with no appreciable time delay, one party might send a "challenge" to which the other entity is the only one who can respond correctly (e.g., challenge for identification).
- **Data Origin Authentication (Delayed):** If messages are exchanged with a delay (stored and forwarded later), it is not convenient to challenge and await a response. Different techniques are required to authenticate the originator of the message, known as data origin authentication.

Difference from Authorization: It is also critical to understand that authentication is done *prior* to the authorization process. In authentication, verification of users is done to determine whether the person is a valid user (requiring login details). In authorization, validation is done to determine if the user has the requisite permission or security level to access specific data.

Question 13: Discuss the core elements/governing principles of data security (CIA Triad: Confidentiality, Integrity, Availability). Explain how confidentiality can be maintained and availability can be ensured in a data management system.

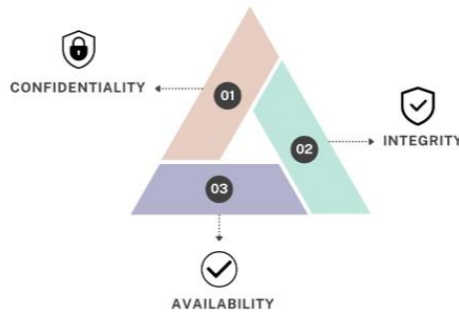
Answer:

Core Elements of Data Security (The CIA Triad): The core elements of data security are confidentiality, integrity, and availability. Also known as the CIA triad, this is a security model and guide for organizations to keep their sensitive data protected from unauthorized access.

The three governing principles are as follows:

- **Confidentiality:** Confidentiality or privacy refers to measures taken to ensure that data—particularly sensitive data—is protected from unauthorized access. Keeping in mind the age of ultra-modern technology, privacy is required to be a basic design consideration. The extent of the level of confidentiality can vary based on the data type and/or regulation.
- **Integrity:** Integrity pertains to safeguarding the accuracy of data as it travels through workflows. There should be measures taken to protect data from unauthorized deletion or modification and to quickly reverse the damage in the event of a breach.
- **Availability:** Availability means providing seamless and continuous access to users through robust servers and network infrastructure with high availability mechanisms built into system design.

CIA Triad



How Confidentiality Can Be Maintained:

Putting confidentiality into practice involves several steps:

- Categorization of data and assets handled by individuals in an organization based on their privacy requirements.
- Requirement of all data encryption and two-factor authentication to be basic security hygiene as a fundamental practice in all organizations dealing with sensitive information.
- Ensure that access control lists and file permissions are monitored and updated regularly by professionals from the IT department in an organization.

How Availability Can Be Ensured:

Ensuring availability involves the following practices:

- Build preventive measures into system designs, make security audits routine, auto-update, or stay alert of system, network, and application updates.
- Utilize detection tools such as network/server monitoring software and anti-virus solutions, and conduct regular check-ups through timely runs.

Question 14: What is the need for security management? Explain the six principles of security management.

Answer:

Need for Security Management: Security management means minimizing the interruption of business activities and reducing the vulnerability to various attacks. Security bargains with distinctive trust aspects of information. Security is not just restricted to computer systems; it applies to all perspectives of securing data or information, in whatever structure. Security is accomplished utilizing a few methodologies at the same time or utilized in blend with one another.

The Six Principles of Security Management:

1. **Availability:** The continuous accessibility of systems tends to procedures, policies, and controls which are used to ensure prompt access to data for authorized customers. This purpose secures against deliberate or inadvertent endeavours to refute legitimate costumers' access to data.
2. **Integrity of data or systems:** System and data integrity is linked to the procedures, policies, and controls which are used to guarantee that data has not been modified in an unconstitutional way and that systems are liberated from illicit manipulation that would compromise precision, comprehensiveness, and consistency.
3. **Confidentiality of data or systems:** Confidentiality covers the procedures, policies, and controls which are utilized to secure data of customers and the organization against illicit access or use.
4. **Accountability:** Accountability incorporates the procedures, policies, and controls essential to follow activities to their source. Accountability specifically underpins non-repudiation, anticipation, infringement, deterrence, security checking, recuperation, and legitimate tolerability of records.
5. **Assurance:** Assurance addresses the procedures, strategies, and controls which are used to create certainty that specialized and equipped security measures are working as anticipated.
6. **Privacy:** It centres on the constitutional rights of people, the motivation behind data assortment and processing, security predilection, and the manner in which organizations administer individual's data. It focuses on how to gather, process, offer, document, and erase the information/data as per the law.

Question 15: Discuss in brief the various data security measures (e.g., E-mail security, Anti-virus protection, Access security).

Answer:

Security threats are inventive and constantly evolve to harm, steal, or corrupt information stored in an organization's system. Organizations must arm themselves with resources to safeguard against these threats. Below are the various crucial data security measures:

- **Access security:** By restricting access to users who have been granted permission, organizations can monitor exactly who has access to particular data. In cases of data theft, sifting through timelines of granted access makes it easier to track down the culprit.
- **Data encryption:** When data is kept unencrypted, it leads to the misuse of personal data by cybercriminals. Data must be encrypted using unique encryption codes to avoid the leakage of vital information. Since only the authorized user has the decryption code, data theft is prevented.
- **Email security:** This is a procedure to protect an email account and its contents from unauthorized access. Measures like strong email passwords and end-to-end encryption prevent the misuse of data, as emails are a popular forum for hackers to spread malware, spam, and phishing attacks.
- **Strong firewall:** A firewall enables protection of a system from internet traffic and exposed services. It enables control over who gains access to an organization's network, effectively stopping insider attacks.
- **Antivirus protection:** Antivirus software is designed to avoid, detect, and deal with cyber security threats. It runs background scans on a system to detect and restrict unauthorized access in the form of malware and protects a system from vulnerabilities.
- **Risk-assessment analysis:** Conducting risk assessments helps an organization understand and assess internal and external risks to the security and confidentiality of personal information stored on various media.
- **Monitor effectiveness:** It is critical to verify security programs through regular tests and monitoring (annually or quarterly) to assess the number of attacks made on an organization's data.
- **Third-party issues:** Organizations must ensure that all reasonable steps have been taken prior to giving access to third-party service providers, and that these providers apply the most stringent security measures.
- **Back-up regularly:** To avoid loss of data due to destruction or unauthorized access, data should regularly be backed up and stored safely.

Question 16: Explain the following terms: (a) Data Management (b) Security audit (c) Security policy.

Answer:

(a) Data Management Data management is the responsible stewardship of data throughout its lifecycle. The main aim of data management is to help people and organizations use data within the boundaries of policies and regulations for the maximum benefit of businesses, making it a highly valuable intangible asset. It can be achieved by the practice of collection, keeping, and usage of data in a secure, efficient, and cost-effective manner. There are five core components to data management:

1. Acquisition
2. Utilization
3. Maintenance
4. Access
5. Protection

(b) Security Audit Security auditing is a vital part of assessing the security strength of data frameworks and systems for any organization. It provides a reasonable and computable way to scrutinize how secure a site really is. Because of its highly specialized and technical nature, security examination is often outsourced to reliable and competent security evaluators. This assessment is specifically designed to:

- Create a security benchmark for the organization.
- Identify the qualities and shortcomings of current security practices.
- Prioritize exposures that present the most serious hazard.
- Provide hazard alleviation proposals consistent with compliance guidelines and industry best practices.

(c) Security Policy A security policy defines the administrative framework and strategies used to protect cyberspace. In 2013, the Government of India formalized this by releasing the National Cyber Security Policy.

- **Purpose & Mission:** The primary purpose of the policy is to create a safe and resilient cyberspace for individuals, organizations, and the government. Its mission is to secure data, develop capacity to avert and react to cyber-attacks, and mitigate harm through institutional collaboration.
- **Strategies Include:** Creating an assertion structure, encouraging open standards, strengthening the administrative structure combined with intermittent audits, and securing e-administration by executing worldwide accepted procedures such as the extensive utilization of Public Key Infrastructure (PKI).

Question 17: What is the need for regulating cyberspace? Explain any two strategies for regulating the Internet.

Answer:

Need for Regulating Cyberspace:

The rapid increase in the use of the online environment has brought numerous legal and social issues that necessitate regulation. The need for regulation arises due to the following reasons:

1. **Distribution of Objectionable Content:** There is a highly visible concern regarding the use of the internet for the distribution of obscene, indecent, and pornographic content, particularly child pornography and child sexual abuse, which calls for strict regulation.
2. **Jurisdiction and Intellectual Property:** Cyberspace poses a challenge to traditional notions of jurisdiction. The shift from tangible assets to intangible assets like Intellectual Property has made cyberspace an important commercial space, requiring regulations to protect these rights.
3. **Invasion of Privacy:** New technologies have enhanced the possibility of invading individual privacy and bringing it into the public domain. Regulation is highly desirable for data protection and data privacy to ensure netizens and industries can have confidence as online users.
4. **Encryption Concerns:** The debate over whether the public should be permitted to use "cryptography" (which can hide communications from third parties) has made regulation important, as new cryptographic techniques can sometimes be cracked or misused.
5. **Lack of Editors:** Unlike traditional press where editors check facts before circulation, the internet allows people to produce and circulate whatever they want. This demands regulation to reconcile freedom of expression with concern for community standards.
6. **Serious Crimes and Anonymity:** Money laundering and other serious crimes become much simpler through the net. Since there are no mechanisms to strictly tie an electronic address to a single identity, the fear of retribution is low, leading to fake identities, rumor-mongering, hate-mail, and virus distribution.

Strategies for Regulating the Internet: According to James Boyle, the regulation of the internet can increasingly rely on specific strategies. Two primary strategies are:

- **Privatization:** The state can use a private body to achieve goals that it could not get directly, and then implement that body's decision through mandatory technological arrangements. For example, the state could make Internet Service Providers (ISPs) strictly liable for copyright violations by their subscribers, thereby creating a private police force with strong incentives to use technical enforcement measures.
- **Technological Controls:** This involves designing the system's architecture to hardwire desired regulatory features. For example, digital texts and music could be encoded to a particular person, detection devices could be built into media players, or blocking software and internet contents grading systems could be mandated by the government.

Question 18: Explain the role and purpose of using filtering devices and rating systems in the context of the legal and self-regulatory framework of cyberspace.

Answer:

In the context of cyberspace regulation, the internet's design precludes centralized control, so outstanding progress has been made in identifying appropriate structures for industry self-regulation with minimal government intervention. Filtering devices and rating systems play a massive role in this self-regulatory framework.

Role of Filtering Devices:

- 'Filters' are software tools specifically used to block access to unwanted material.
- For example, e-mail filters automatically delete the bulk of unread messages commonly known as 'spam' and can be customized to block incoming messages from particular sources.
- Site blocking filters screen out specified websites or websites containing specific keywords that the system presumes relate to objectionable content.

Role of Rating Systems:

- Rating systems classify the content on an Internet site. For instance, the Platform for Internet Content Selection (PICS) was developed by the World Wide Web Consortium (W3C) to create common protocols for the web.
- A prominent example is the RSACI (Recreational Software Advisory Council labeling scheme for the Internet) rating system, which addresses the level of violence, sex, nudity, and language on a website.
- It operates as a classification tool rather than making a judgment about the content's appropriateness for any given audience, which allows parents and schools to make their own choices.

Purpose in the Self-Regulatory Framework: The primary purpose of developing technology to permit content labeling, rating systems, and filtering tools is to provide the necessary ingredients for self-regulatory schemes.

- It empowers the end-user through the voluntary use of filtering/blocking technologies to protect themselves and minors from problematic content.
- Rating systems (like RSACI) have a distinct advantage over keyword-based filtering programs. Keyword filters often inevitably block a significant amount of useful, inoffensive content simply because a "bad" word was detected. Rating systems solve this by looking at the labeled classification of the site instead.

Question 19: Explain the UNCITRAL Model Law, 1996.

Answer:

What is the UNCITRAL Model Law? The United Nations Commission on International Trade Law (UNCITRAL) is a Model Law on Electronic Commerce that was adopted by the UN Commission in 1996. Its specific goal is to eliminate all kinds of legal and technical hindrances and to boost legal certainty for the cyber domain by postulating a series of internationally acceptable rules.

It is important to note that the Information Technology Act, 2000 of India was enacted on the exact same grounds as this model law.

The Three Core Doctrines:

The Model Law has embraced the following three fundamental principles:

1. **Non-discrimination principle:** No document can be denied legal enforceability simply because it exists in an electronic means.
2. **Technological neutrality principle:** It adapts to existing laws which are neutral in nature so that the same can be made applicable to any future technological innovations.
3. **Equivalence functional principle:** It affords uniform treatment to transactions done online exactly as that of transactions done offline.

Structure and Provisions of the Model Law: This Model Law has been divided into two parts. Part 1 deals with general provisions regarding e-commerce, and Part 2 deals with specific provisions for e-commerce in certain areas.

- **Article 1:** Deals with information in the form of data messages within the framework of commercial activities.
- **Article 2:** Covers definitions, such as "data message," which uses the phrase 'similar means' to ensure future developments are included.
- **Article 3 & 4:** Cover interpretation techniques and the variation in communication by agreement between both parties.
- **Article 5:** States that information in electronic mode cannot be denied legal validity or enforcement.
- **Article 6 & 7:** Removes the traditional hindrances of requiring a document to be "in writing" and having "handwritten signatures".
- **Article 8:** Talks about the standard of reliability (the reason the data was generated and surrounding circumstances).
- **Articles 9 to 15 (General):** Discuss the evidentiary value granted to messages stored online, the retention of information, the formation of valid contracts, and acknowledgment in the form of receipts.
- **Article 16 & 17 (Specific):** Enlist rules for the carriage of goods and transport documents.

Question 20: Discuss the laws in the United States and the United Kingdom for the regulation of cyberspace.

Answer:

Laws in the UNITED STATES The exponential growth in the usage of online services in the US led to demands for its operations to be regulated. The major legislations include:

- **Communications Decency Act 1996 (CDA):** It prohibited the making and transmission of obscene or 'indecent' material to a minor by means of a telecommunications device. (Note: Supreme Court later declared some provisions unconstitutional for violating freedom of speech)
- **Children Online Privacy Protection Act (COPPA), 1998:** Applies to operators of commercial websites directed to children under 13 that collect, use, or disclose personal information from children.
- **Children Internet Protection Act (CIPA), 2000:** Requires schools and libraries to install filters on computers used by minors and adults.
- **Uniform Electronic Transactions Act (UETA), 1999:** Aims to remove barriers to electronic commerce by validating electronic records and signatures.
- **Uniform Computer Information Transaction Act (UCITA), 2000:** Applies to contracts for the development or creation of computer information, such as software development contracts and databases.
- **Other Related Laws:** The Gramm-Leach-Bliley Act (GLBA) clarifies how financial institutions share confidential info , and the Health Insurance Portability and Accountability Act (HIPAA) protects patient's sensitive health information.
- **Cyber Security Specific Laws:** Include the Consumer Privacy Protection Act 2017 , Cybersecurity Information Sharing Act (CISA) , and the Cybersecurity Enhancement Act of 2014.

Laws in the UNITED KINGDOM

The UK relies heavily on both self-regulation and specific legislations to manage cyberspace.

- **Internet Watch Foundation (IWF):** Originally the R3 Safety-Net action plan, this was developed by UK ISP trade associations to establish a complaints hotline and take-down procedures for illegal Internet content, primarily child pornography and "criminally racist content".
- **The Computer Misuse Act, 1990:** Makes provision for securing computer material against unauthorized access or modification.
- **Electronic Communications Act, 2000:** Created to facilitate the use of electronic communications and electronic data storage.
- **Data Protection Act 2018 (DPA 2018):** Superseded the Data Protection Act 1998 and supplements the EU General Data Protection Regulation (GDPR). It regulates the processing of information relating to individuals.
- **National Cyber Security Centre (NCSC):** Formed in 2016 by merging various security bodies. It provides a single point of contact for SMEs, larger organizations, and government agencies, working in collaboration with intelligence agencies to assess and protect national infrastructure.

Question 21: Give an overview of Cyberspace regulation in India.

Answer:

Overview of Cyberspace Regulation in India: In India, the **Information Technology Act, 2000** is the primary legislation which covers the domain of cyber law. The main objective of the Act is to provide legal recognition for transactions carried out by means of electronic data interchange and other means of electronic communication (commonly referred to as e-commerce), which involve the use of alternatives to paper-based methods of communication. It also aims at facilitating the electronic filing of documents with Government agencies.

Key Provisions and Chapters under the IT Act, 2000:

- **Electronic Signatures (Chapter II):** Any subscriber may authenticate an electronic record by affixing their Digital Signature. Electronic records include data, images, or sound stored or received in an electronic form.
- **Electronic Governance (Chapter III):** Where any law requires information to be in writing or printed, it is legally sufficient if it is sent in an electronic form. Furthermore, the filing of forms, applications, or the issue of licenses and receipts by Government offices can be done electronically.
- **Regulation of Certifying Authorities (Chapter IV):** The Central Government appoints a Controller of Certifying Authority to exercise supervision over the activities of Certifying Authorities.
- **Digital Signature Certificate (Chapter VII):** Any person may apply to the Certifying Authority to issue a Digital Signature Certificate.
- **Penalties and Adjudication (Chapter IX):** If any person, without permission, accesses a computer/network, downloads extracts, introduces viruses, or damages data, they shall be liable to pay damages by way of compensation not exceeding Rupees One Crore to the affected person.

- **The Appellate Tribunal (Chapter X):** The Telecom Disputes Settlement and Appellate Tribunal acts as the Appellate Tribunal for this Act. It entertains cases of any person aggrieved by an Order made by the Controller or the Adjudicating Officer.
- **Offences (Chapter XI):** Tampering with computer source documents or hacking entails punishment with imprisonment up to three years or a fine up to Rs. 2 lakhs, or both. Publishing obscene information electronically is punishable with imprisonment up to five years or a fine up to Rs. 10 lakhs for the first conviction.
- **Important Amendment (2015):** The Supreme Court of India in the landmark case of *Shreya Singhal v. Union of India* struck down Section 66A of the IT Act, 2000, as it violated the freedom of speech and expression guaranteed under the Constitution of India.

Question 22: How is cybercrime different from traditional crime? List the four categories of cybercrimes and list various cybercrimes under each category.

Answer:

How Cybercrime is Different from Traditional Crime: Cybercrime is defined as crimes committed on the internet using the computer as a tool to target the victim. Cybercrimes are quite different from traditional crimes for the following reasons:

1. **Difficulty of Detection:** They are often much harder to detect, investigate, and prosecute compared to traditional crimes. Because of this, cybercrimes cause greater damage to society.
2. **Lack of Physical Evidence:** In traditional crimes, criminals usually leave physical proof like fingerprints. In cybercrimes, criminals operate through the internet, leaving very few chances of leaving any physical proof behind.
3. **Distance and Anonymity:** It is difficult to determine where the particular cybercrime took place because it can harm its victim even when the attacker is sitting at a far distance. (*Note: Traditional crimes like defamation, forgery, identity theft, terrorism, and bullying are also considered cybercrimes when they are conducted through the internet or computer technology.*)

Four Categories of Cybercrimes: Cybercrimes are broadly classified into four main groups:

1. **Crime against the individuals:** Includes harassment, cyber-stalking, defamation, indecent exposure, cheating, email spoofing, and fraud.
2. **Crime against property:** Includes transmitting viruses, net-trespass, unauthorized control over a computer system, internet thefts, and infringement of intellectual property.
3. **Crime against organization:** Includes cyber terrorism within a government organization, possession of unauthorized information, and distribution of pirated software.
4. **Crime against society:** Includes child pornography, financial crimes, sale of unlawful articles, trafficking, forgery of records, and gambling.

Question 23: Discuss the acts/circumstances under which a person (wrongdoer) is liable to pay damages/compensation under Section 43 of the Information Technology Act, 2000.

Answer:

Liability to Pay Damages/Compensation under Section 43: Section 43 of the Information Technology Act, 2000 provides that if any person, without the permission of the owner or any other person who is in charge of a computer, computer system, or computer network, commits specific wrongful acts, they shall be liable to pay damages by way of compensation to the affected party.

The specific acts and circumstances under which a wrongdoer becomes liable are:

1. **Unauthorized Access:** Accesses or secures access to such computer, computer system, or computer network.
2. **Data Extraction:** Downloads, copies, or extracts any data, computer database, or information from such computer.
3. **Introducing Malware:** Introduces or causes to be introduced any computer contaminant or computer virus into any computer system or network.
4. **Causing Damage:** Damages or causes to be damaged any data, computer database, or any other programs residing in such computer.
5. **Causing Disruption:** Disrupts or causes disruption of any computer, computer system, or computer network.
6. **Denial of Access:** Denies or causes the denial of access to any person authorized to access any computer or computer network by any means.
7. **Facilitating Unauthorized Access:** Provides any assistance to any person to facilitate access to a computer in contravention of the provisions of this Act.

8. **Account Manipulation:** Charges the services availed of by a person to the account of another person by tampering with or manipulating any computer system.
9. **Altering Information:** Destroys, deletes, or alters any information residing in a computer resource, diminishes its value or utility, or affects it injuriously by any means.
10. **Stealing Source Code:** Steals, conceals, destroys, or alters (or causes any person to do so) any computer source code used for a computer resource with an intention to cause damage.

Question 24: Discuss the offences in the context of the regulation of cyber law content in India.

Answer:

Offences under the Information Technology Act, 2000: While Section 43 deals with civil liability (compensation), the IT Act also specifies various acts as criminal offences which attract imprisonment and fines. The key offences in the context of regulating cyber law content in India include:

1. **Tampering with Computer Source Documents (Section 65):** Intentionally concealing, destroying, or altering any computer source document is an offence. Penalty: Imprisonment up to 3 years, or a fine up to Rs. 2,00,000, or both.
2. **Computer Related Offences (Section 66):** Dishonestly or fraudulently doing any act referred to in Section 43 (like unauthorized access or introducing viruses) is an offence. Penalty: Imprisonment up to 3 years, or a fine up to Rs. 5,00,000, or both. (*Note: Section 66A was struck down by the Supreme Court in 2015 as it violated freedom of speech*).
3. **Receiving Stolen Computer Resource (Section 66B):** Dishonestly or fraudulently receiving or retaining any stolen computer resource or communication device. Penalty: Imprisonment up to 3 years, or a fine up to Rs. 1,00,000, or both.
4. **Identity Theft (Section 66C):** Dishonestly or fraudulently making use of the Electronic Signature, Password, or any other Unique Identification Feature of any other person. Penalty: Imprisonment up to 3 years, or a fine up to Rs. 1,00,000, or both.
5. **Cheating by Personation (Section 66D):** Cheating by personating using any communication device or computer resource. Penalty: Imprisonment up to 3 years, or a fine up to Rs. 1,00,000, or both.
6. **Violation of Privacy (Section 66E):** Intentionally capturing, publishing, or transmitting the image of the private area of any person without their consent. Penalty: Imprisonment up to 3 years, or a fine up to Rs. 2,00,000, or both.
7. **Cyber Terrorism (Section 66F):** Doing any act electronically or with the use of a computer with the intent to threaten the unity, integrity, security, or sovereignty of India. Punishment: Imprisonment for Life.
8. **Publishing Obscene Material (Section 67):** Publishing or transmitting material in electronic form which appeals to the prurient interest or tends to deprave and corrupt persons. Penalty: Imprisonment up to 3 years and a fine up to Rs. 5,00,000 (increases for subsequent convictions).
9. **Publishing Sexually Explicit Material (Section 67A):** Publishing or transmitting material which contains a sexually explicit act or conduct. Penalty: Imprisonment up to 5 years and a fine up to Rs. 10,00,000.

Question 25: What are the liabilities of 'Network Service Providers' (intermediaries) in the context of cybercrimes? Explain the grounds for their exemption.

Answer:

Who are Network Service Providers? A 'network service provider' (intermediary) means any person who provides access to an information service in electronic form. Examples include Internet Service Providers (ISPs), cellular mobile services, customer access services, and mobile satellite services. It essentially performs two tasks: providing access to the network and acting as an intermediary between an originator and addressee with respect to any particular electronic message.

Grounds for Exemption from Liability (Section 79): Section 79 of the Information Technology Act provides certain exemptions from liabilities to intermediaries. An intermediary shall not be liable for any third-party information, data, or communication link made available or hosted by him if the following conditions are met:

1. **Limited Function:** The function of the intermediary is limited strictly to providing access to a communication system over which information made available by third parties is transmitted, temporarily stored, or hosted.
2. **No Direct Involvement:** The intermediary does not initiate the transmission, select the receiver of the transmission, and select or modify the information contained in the transmission.
3. **Due Diligence:** The intermediary observes due diligence while discharging his duties under the Act and also observes such other guidelines as the Central Government may prescribe in this behalf.

When Does the Exemption NOT Apply? The safe harbor exemption will be revoked, and the intermediary will be held liable if:

1. **Conspiracy:** The intermediary has conspired, abetted, aided, or induced, whether by threats or promise or otherwise, in the commission of the unlawful act.

2. **Failure to Remove Unlawful Content:** Upon receiving actual knowledge, or on being notified by the appropriate Government or its agency that any information residing in a computer resource controlled by the intermediary is being used to commit an unlawful act, the intermediary fails to expeditiously remove or disable access to that material on that resource without vitiating the evidence in any manner.

Question 26: What is Cyber Forensics? Explain the need and importance of the preservation of Digital Evidence and list Cyber Forensic Investigation tools.

Answer:

What is Cyber Forensics? Cyber forensics (also known as computer forensics or digital forensics) is the application of investigation and analysis techniques to gather and store evidence from a particular computing system in a way that is appropriate for presentation in a court of law. It involves the acquisition, authentication, analysis, and documentation of evidence retrieved from the systems (computers, networks, digital media) or online domains used to commit the crime.

Need and Importance of Preserving Digital Evidence:

1. **Fragility:** Digital evidence is highly fragile; it can be easily damaged, modified, or destroyed purposely. Therefore, the original evidence is often duplicated, and analysis is carried out on the duplicated copy (via Imaging) to prevent any mishap of damaging the original.
2. **Uncovering Truth:** Computer forensics is important because it recovers hidden or deleted files from storage devices and tells experts whether there are any suspicious activities going on or if systems have been tampered with.
3. **Chain of Custody:** The preservation of digital evidence is crucial to maintain an unbroken "chain of custody." This acts as a map depicting how digital evidence was processed, collected, analyzed, and preserved. If the chain of custody is broken, the evidence submitted to the court can be denied because it might have been altered and might not tell the truth.
4. **Write-Blocking:** Investigators use methods like 'Write-block' connecting evidence media with a special machine to prevent any attempt to overwrite the data on the device, ensuring it stays in its original state.

Cyber Forensic Investigation Techniques and Tools: The common techniques used by forensic experts include:

- **Cross-driven analysis:** Correlates data from multiple hard drives.
- **Live analysis:** Obtains data acquisitions before a PC is shut down.
- **Deleted file recovery:** Extracting digital evidence from the source or hard drive.
- **Stochastic Forensics:** Used for detecting data theft.
- **Steganography:** Analysis of concealed files, messages, images, or videos within another file.

Forensic Tools: Investigators often use specific tools like **OS forensics, XYR tools, and Quickstego** or other sophisticated toolkits to aid them in finding and collecting evidence, uncovering hidden files, and determining relevance to the case.

Question 27: What is the meaning of Intellectual Property? Describe in brief the various forms of Intellectual Property (e.g., Copyright, Patents, Trademarks, etc.).

Answer:

Meaning of Intellectual Property: The term Intellectual Property (IP) can be defined as intangible property which is the creation of one's mind. It is not merely an idea but an expression of it, such as musical, literary and artistic works, inventions, designs, symbols, names, and images. The main ingredient for fulfilling the condition of intellectual property is "novelty". The rationale behind providing IP rights is to give creators due recognition and monetary benefits, which encourages further innovation and economic growth.

Various Forms of Intellectual Property: There are basically seven forms of intellectual property:

1. **Copyright and related rights:** A copyright protects creative literary, musical, dramatic, or artistic works like cinematograph films, sound recordings, paintings, books, and computer software. It grants the owner exclusive rights to reproduce, distribute, broadcast, and publicly perform the work.
2. **Patents:** An exclusive right granted to protect an invention (a product or process) that is novel, involves an inventive step, and is capable of industrial application. It is territorial in nature and granted for a limited period.
3. **Trademarks:** A distinctive sign, word, symbol, logo, or mark used in trade to distinguish goods or services. It helps consumers identify the source of products or services. Registration is valid for a specific period and needs to be renewed.
4. **Industrial Designs:** Protects the appearance of a product/logo, the shape of an object, composition of design, or pattern of cloth. It must be new, original, and aesthetic, requiring mandatory registration.

5. **Geographical Indications (GI):** A name or sign used on certain products which corresponds to a specific geographical location or origin (e.g., Basmati rice, Swiss watches). The product's reputation or quality must essentially be due to its place of origin.
6. **Trade Secrets:** Confidential secret information having an inherent economic advantage to a company (e.g., formulae, recipes, business techniques). No registration is required, but businesses must actively limit access and sign non-disclosure agreements.
7. **Plant Variety:** A form of IP protecting newly engineered plant species or strains.

Question 28: Critically examine the copyright issues in the digital medium. Briefly explain the Universal Copyright Convention.

Answer:

Copyright Issues in the Digital Medium: The global usage of the Internet makes it feasible for any user to easily share information in cyberspace, which presents a troublesome situation for copyright holders. The primary issues include:

1. **Mass Dissemination & Piracy:** The Internet allows users to become mass disseminators of others' copyrighted material, creating a disequilibrium between authors and users and leading to huge monetary losses through piracy.
2. **Reproduction Right Violations:** The most fundamental right of a copyright owner is controlling the reproduction of their work. Digital technology and networks allow for easy copying and transmission, making the copyrighted work less distinct and leading to widespread infringement.
3. **Repackaging and Retransmitting (The "Fair Use" Defense Failure):** Issues arise when platforms copy materials (like music CDs) onto their servers merely to repackage and retransmit them without authorization. In cases like *UMG Recordings, Inc. v. MP3.Com, Inc.*, the court explicitly ruled that copying purchased CDs onto servers to replay the recordings for subscribers over the internet is *not* a "fair use" and strictly constitutes copyright infringement.

The Universal Copyright Convention (UCC): The Universal Copyright Convention (UCC) is an important international treaty adopted in 1952 under the support and protection of the United Nations Educational, Scientific and Cultural Organization (UNESCO).

- **Purpose:** Its main purpose is to extend international copyright protection universally and create uniformity in dealing with disputes related to IP rights across member countries.
- **Revisions and Compliance:** It was revised in 1971, and members must strictly comply in accordance with this revised version.
- **Administration:** An Intergovernmental Copyright Committee was established in compliance with Article 11 of the UCC, consisting of representatives of 18 Contracting States to oversee its implementation.

Question 29: Discuss the issues arising due to Linking, Inlining, and Framing. What is framing? Explain the liability for framing with the help of case laws.

Answer:

Issues Arising due to Linking, Inlining, and Framing: Linking, inlining, and framing have become very common practices on the internet, but they raise several legal concerns. The primary issues that arise from these practices include:

1. **Loss of Income:** In linking, a person provides a link that allows visitors to bypass information and advertisements at the original home page, causing a direct loss of income to the original businesses.
2. **Confusion and False Endorsement:** These practices can create confusion among users, making them believe that the sites endorse each other or are officially associated with each other, which might not be correct.
3. **Loss of Reputation:** It can lead to confusion regarding the original source of the content, resulting in a loss of reputation and goodwill for the original information holder or business.
4. **Copyright and Trademark Infringement:** Pulling full-size images or displaying graphics originating from another website can trigger disputes under copyright and trademark laws.

What is Framing? "Framing" is the process of allowing a user to view the contents of one website while it is framed by information from another site, similar to the "picture-in-picture" feature offered on some televisions. Framing may trigger a dispute under copyright and trademark law theories because a framed site arguably alters the appearance of the original content and creates the false impression that its owner endorses or voluntarily chooses to associate with the framer.

Liability for Framing (With Case Law): The liability for framing generally falls under the infringement of creating an unauthorized "derivative work".

- **Case Law - *Futuredontics Inc. v. Applied Anagramic Inc.*, 2007:** In this case, a district court ruled that the addition of reproduced Web pages within a "frame" by a dental website (containing contents of another website detailing Applied Anagramic, its trademarks, and links) leads to a modification in the appearance of the linked site. The court held that such modifications could, without authorization, amount to the infringement of a derivative work. (*Note: To avoid liability, one must seek permission from the original owner or clearly display a disclaimer stating the source of the information.*)

Question 30: What is a Trademark? What are the possible trademark infringements related to search engines?

Answer:

What is a Trademark? A trademark is a distinctive sign, word, symbol, or mark used in trade to distinguish the goods or services of one enterprise from those of others. Trademarks help consumers identify the source of products or services. It could be a name, signature, logo, brand label, phrase, slogan, letter, a numeral, or any combination of them.

Trademark Infringements Related to Search Engines: Search engines connect users with the World Wide Web, but there is a possibility of unfairness in this process that leads to trademark infringement issues. The possible infringements include:

1. **Sponsored Ads and Keyword Purchasing:** The main source of income for search engines is advertisements shown alongside search content. Search engines allow advertisers to purchase advertising space on the page of a trademark actually searched online by users. Trademark owners object to this practice of showing sponsored ads of competitors on their trademarked keywords, considering it a violation of their trademark in cyberspace, an unfair trade practice, and highly misleading. It has the potential to create confusion in the mind of the consumer regarding the trademark or keyword searched.
2. **Abuse of Meta Tags:** Meta tags are hidden text used by search engines to scan and know about a webpage. Although they do not affect the visible appearance of a website, companies can use them in a deceptive manner by burying misleading terms or a competitor's trademark in their hidden text. This diverts or confuses e-consumers. For example, a shoe manufacturing company may bury the meta tag "Bata" in its Web page to lure internet users who are actually searching for genuine Bata products.

Question 31: What is meant by Domain Name disputes (Cybersquatting)?

Answer:

What is a Domain Name? A domain name is the internet/web address of a website and is a component of a URL (Uniform Resource Locator) which makes it easy to identify the Internet protocol or IP address. Domain names and trademarks represent prominent marketing tools and the identity of a business, carrying the goodwill and reputation attached to it.

What is Cybersquatting (Domain Name Disputes)? Cybersquatting, also known as "domain name hijacking", is a form of domain name misuse. It constitutes the act of registering a domain name with *malafide* (bad faith) intention which is actually someone else's trademark.

- **The Motive:** People create and register domain names of other real owners as their own and take advantage of it by selling them back to the real trade owner at an excessive price.
- **The Impact:** It is an unscrupulous practice that leads to misrepresentation in the eyes of potential buyers or service users, impacting global trade and infringing the rights of a particular trademark.

Case Law Example: In the famous case of *Yahoo! Inc. v. Akash Arora*, 1999, the defendant created and registered a deceptively similar website domain name "YahooIndia.com" and started providing similar services. The High Court of Delhi held that the defendant was liable for deceptively using Yahoo as a domain name and passed an injunction order to restrain the misuse.

Question 32: Explain Patent misuse.

Answer:

What is Patent Misuse? A patent provides a monopoly to the patentee, meaning they have the exclusive right to present their knowledge to the public, and the invention cannot be commercially used, made, distributed, or sold without their consent.

However, a patent can also be misused by the patent holder. Patent misuse refers to the illegal behavior of a patentee that leads to violations of antitrust laws, or when the patentee tries to inappropriately expand their product with the actual patent by getting into restrictive licensing agreements.

Consequence of Misuse: When patent misuse has been constituted, the patent is deemed useless by the courts.

Examples in the Digital/Technology Industry: With the advent of digitized media, various patent infringements and misuses have been seen in technology industries:

1. **Amazon:** Tried to patent its "one-click" payment option, but the court decided it was too obvious an idea to patent.
2. **Napster:** The file-sharing company settled a lawsuit accusing it of unauthorized distribution of music and later filed for bankruptcy.
3. **Nintendo:** Was forced to pay a large sum to Tomita Technologies for its 3DS gaming-system technology.
4. **Microsoft vs. Google:** Dueled for five years over patent issues involving the Xbox gaming system and Motorola smartphones.

Question 33: What are the different types of remedies available against the infringement of IPR (Intellectual Property Rights) / Trademarks?

Answer:

According to the laws regulating Intellectual Property in India, there are three primary types of remedies available to an owner against the infringement of their IPR (such as Trademarks and Copyrights):

1. **Civil Remedies:** These are remedies sought in a civil court to stop the infringement and recover losses. They include:
 - Injunctions (court orders to stop the infringing activity).
 - Damages (monetary compensation for losses).
 - Rendition of accounts (handing over the profits made from the infringement).
 - Ex-parte orders (orders passed without the other party present).
 - Seizure, destruction, or forfeiture of the infringing goods.
2. **Criminal Remedies:** These involve penalizing the infringer with imprisonment and fines:
 - Section 63 of the Copyright Act, 1957 deals with the offences of infringement of copyright.
 - Chapter XII of the Trademarks Act, 1999 deals with offences, penalties, and procedures pertaining to trademark infringement.
3. **Administrative Remedies:** These involve actions taken by government authorities (like customs) to prevent the circulation of infringing goods:
 - Checking the import/export of goods, including the protection of patents, trademarks, and copyrights under the Indian Customs Act, 1962.
 - Confiscation of infringing material by Custom Authorities.
 - Restrictions against the parallel importation of goods.

Question 34: Discuss the acts/circumstances under which a person is liable to pay damages/compensation under Section 43 of the Information Technology Act, 2000.

Answer:

Overview of Section 43 (Liability for Compensation): Section 43 of the Information Technology Act, 2000 explicitly deals with civil liability. It states that if any person, without the permission of the owner or the person in charge of a computer, computer system, or computer network, commits specific unauthorized acts, they shall be liable to pay damages by way of compensation to the affected party.

The 10 Acts/Circumstances Attracting Liability: A wrongdoer is liable to pay compensation if they commit any of the following acts without permission:

1. **Unauthorized Access:** Accesses or secures access to such computer, computer system, or computer network.
2. **Data Extraction:** Downloads, copies, or extracts any data, computer database, or information.
3. **Introducing Malware:** Introduces or causes to be introduced any "computer contaminant" or "computer virus". (A contaminant modifies, destroys, or transmits data, while a virus degrades or adversely affects performance).
4. **Causing Damage:** Damages or causes to be damaged any data, computer database, or any other programs.
5. **Causing Disruption:** Disrupts or causes disruption of any computer, computer system, or network.
6. **Denial of Access:** Denies or causes the denial of access to any authorized person.
7. **Facilitating Unauthorized Access:** Provides any assistance to any person to facilitate access to a computer in contravention of the Act.
8. **Account Manipulation:** Charges the services availed of by one person to the account of another person by tampering or manipulating.

9. **Altering Information:** Destroys, deletes, or alters any information residing in a computer resource or diminishes its value/utility.
10. **Stealing Source Code:** Steals, conceals, destroys, or alters any computer source code used for a computer resource with an intention to cause damage.

Question 35: What is Cyber Forensics? Explain the need and importance of the preservation of Digital Evidence and describe Cyber Forensic Investigation tools.

Answer:

What is Cyber Forensics? Cyber forensics (also known as computer or digital forensics) is the application of investigation and analysis techniques to gather and store evidence from a particular computing system in a way that is appropriate for presentation in a court of law. It involves the acquisition, authentication, analysis, and documentation of digital evidence retrieved from computers, networks, storage media, or online domains used to commit a crime.

Need and Importance of the Preservation of Digital Evidence:

1. **Fragility of Data:** Digital evidence is extremely fragile and can be easily damaged, modified, or destroyed purposely by the suspect.
2. **Uncovering Hidden Activity:** It helps forensic experts recover files that are hidden, deleted, or formatted, revealing the true activities of the user.
3. **Maintaining the "Chain of Custody":** When submitting digital evidence for legal purposes, legitimate proof is required to show that the evidence is exactly the same as the exhibit seized at the crime scene. This map of how evidence was processed is called the "chain of custody". If this chain is broken, the evidence can be denied in court because it might have been altered.
4. **Safe Analysis (Imaging & Write-Blocking):** To prevent altering the original evidence, experts create a "ghosted" backup copy (Imaging) to work on. They also use a special machine called a "Write-block" that prevents any attempt to overwrite the original data on the device.

Cyber Forensic Investigation Techniques and Tools: Forensic experts browse through thousands of files, focusing on areas of interest like the Recycle Bin, Windows Registry, and Internet Temp Folders. To do this, they use specific techniques and tools:

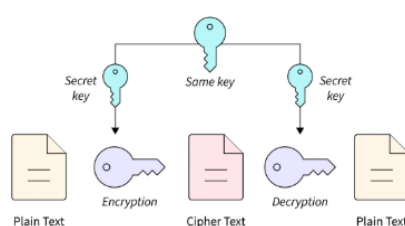
- **Standard Techniques:** * *Cross-driven analysis:* Correlates data from multiple hard drives.
 - *Live analysis:* Obtains data acquisitions before a PC is shut down.
 - *Stochastic Forensics:* Used for detecting data theft.
 - *Steganography:* Finding a file, message, or image concealed within another file.
- **Forensic Software Tools:** Investigators often use specific tools like **OS forensics**, **XYR tools**, and **Quickstego** or other sophisticated toolkits to aid them in finding and collecting evidence safely and efficiently.

Question 36: What do you mean by Encryption? Distinguish between Symmetric key and Public key encryption schemes, mentioning their advantages and disadvantages.

Answer:

What is Encryption? Encryption is a common method of protecting information transmitted over unreliable links. The mechanism works in the following steps:

- The information is encrypted (encoded) from its initial readable form (called clear text) into an internal, unreadable form (called cipher text).
- The cipher text can then be stored in a readable file or safely transmitted over unprotected channels.
- Finally, the receiver must decrypt (decode) the cipher text back into clear text to understand its meaning.



Distinction, Advantages, and Disadvantages:

1. Symmetric Key Cryptography (Private-Key) In this scheme, one secret key is shared and used for both encryption and decryption. The sender and receiver must have a pre-shared key, kept undisclosed from all other parties.

- **Advantages:**
 - **High Speed:** Symmetric-key ciphers have high rates of data throughput and are very fast.
 - **Shorter Keys:** The keys used are relatively short.
 - **Strong Ciphers:** Simple transformations can be composed to produce stronger product ciphers.
- **Disadvantages:**
 - **Secrecy Requirement:** The key must remain an absolute secret at both ends.
 - **Key Management:** Managing many key pairs in a large network requires an unconditionally trusted Third Party (TTP).
 - **Frequent Changes:** Keys must be changed frequently, often for each communication session.

2. Public Key Cryptography (Asymmetric-Key) In this scheme, two different keys are used: a public key for encryption and a private key for decryption. The sender uses the public key, whereas the receiver uses the private key (which is kept secret).

- **Advantages:**
 - **Simplified Secrecy:** Only the private key must be kept secret.
 - **Easier Key Administration:** The administration of keys requires only a functionally trusted TTP, and secret keys don't have to be transmitted across the network.
 - **Long-term Validity:** A private/public key pair may remain unchanged for considerable periods of time.
- **Disadvantages:**
 - **Slower Throughput:** Throughput rates are several orders of magnitude slower than symmetric-key schemes.
 - **Larger Key Sizes:** Key sizes and the size of public-key signatures are typically much larger.

Question 37: Discuss the core elements of Data Security (The CIA Triad). How can these principles be practically implemented in a data management system?

Answer:

Core Elements of Data Security (The CIA Triad): The core elements of data security are confidentiality, integrity, and availability. Also known as the CIA triad, this is a security model and guide for organizations to keep their sensitive data protected from unauthorized access.

1. **Confidentiality:** Refers to measures taken to ensure that data—particularly sensitive data—is protected from unauthorized access. The level of confidentiality can vary based on the data type or regulation.
2. **Integrity:** Pertains to safeguarding the accuracy of data as it travels through workflows. It involves protecting data from unauthorized deletion or modification and quickly reversing damage in the event of a breach.
3. **Availability:** Means providing seamless and continuous access to users through robust servers and network infrastructure with high availability mechanisms built into the system design.

CIA Triad



Practical Implementation in a Data Management System:

- **Putting Confidentiality into Practice:**
 - Categorize data and assets handled by individuals based on their privacy requirements.
 - Require all data encryption and two-factor authentication as a fundamental practice.
 - Ensure that access control lists and file permissions are monitored and updated regularly by IT professionals.
- **Scoping Integrity:**
 - Review all data processing, transfer, and storage mechanisms, and run diagnostic tests to ensure there is no unauthorized access.

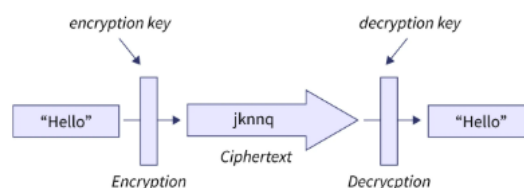
- Understand compliance and regulatory requirements.
- Invest in a dependable backup and recovery solution to ensure quick data recovery in the event of a breach.
- **Ensuring Availability:**
 - Build preventive measures into system designs, make security audits routine, and auto-update systems and applications.
 - Utilize detection tools such as network/server monitoring software and anti-virus solutions, conducting regular check-ups.

Question 38: What is meant by Cryptography? Describe the RSA algorithm with a suitable example.

Answer:

What is Cryptography?

Cryptography is the art of devising ciphers to secure computer systems by encrypting sensitive records and messages in transit and in storage. In cryptography, the message or data to be encrypted (known as the plaintext) is transformed by a function that is parameterized by a "KEY". The output of the encryption process is known as the cipher text, which is then safely transmitted through an insecure communication channel.



The RSA Algorithm:

RSA gets its name from the initials of its three discoverers (Rivest, Shamir, and Adleman) at M.I.T.. It is an asymmetric (public-key) cryptography algorithm because it uses two different keys for encoding and decoding. The entire algorithm is based on the modular mathematics of Number theory.

Steps of the RSA Algorithm:

The determination of the variables for key generation is prescribed in the following mathematical way:

1. **Select Primes:** Choose two large prime numbers, p and q , each typically greater than 10^{100} .
2. **Compute Modulus (n) and Totient ($\phi(n)$):** * Calculate $n = p \times q$.
 - Calculate $\phi(n) = (p - 1)(q - 1)$.
3. **Choose Private Key (d):** Assume a number d to be a large, random integer that is relatively prime to $\phi(n)$.
4. **Choose Public Key (e):** Calculate e such that $e \times d \pmod{\phi(n)} = 1$.
5. **Key Pairs:** The public key is (e, n) and the private decryption key is (d, n) .

Encryption and Decryption Example:

To encipher a plaintext message P (where $0 \leq P < n$):

- **Encryption:** The Cipher text (C) is obtained using the public key: $C = P^e \pmod{n}$.
- **Decryption:** The Cipher text (C) is decrypted back to plaintext using the private key: $P = C^d \pmod{n}$.

These steps ensure that encryption and decryption are exact inverses of each other, making the system highly secure.

Question 39: Discuss the issues arising due to Linking, Inlining, and Framing in Cyberspace. Explain the liability for framing using relevant case laws.

Answer:

Issues Arising Due to Linking, Inlining, and Framing:

These are common practices for sharing information on the World Wide Web, but they raise significant Intellectual Property (IP) issues:

1. **Linking (Deep Linking):** "Linking" allows a website user to visit another location on the internet by clicking a "live" word or image. The issue arises with "deep linking," where a user links directly to a subordinate page of another site. This allows visitors to bypass the original site's homepage and its advertisements, causing a loss of revenue for the original owner.
2. **Inlining:** "Inlining" is the process of displaying a graphic file on one website that actually originates at another. For example, displaying a "cartoon of the day" from Site B seamlessly onto Site A. This can trigger copyright infringement as it reproduces copyrighted imagery without permission.
3. **Framing:** "Framing" is the process of allowing a user to view the contents of one website while it is framed by information (like logos and menus) from another site, similar to a "picture-in-picture" feature.

Liability for Framing (With Case Law):

Framing triggers severe disputes under copyright and trademark law because a framed site alters the appearance of the original content and creates a false impression that the original owner endorses or is associated with the framer.

- **Case Law - *Futuredontics Inc. v. Applied Anagramic Inc. (1997/2007)*:** In this case, a district court ruled on a dental website containing the contents of another website within a "frame" that featured the framer's trademark and links. The court ruled that the addition of the reproduced Web pages within a frame leads to a modification in the appearance of the linked site. Such modifications, done without authorization, could amount to the infringement of a "derivative work" under copyright law.

To avoid liability, a website must seek explicit permission or use a prominently displayed disclaimer stating the source of the information.

Question 40: How is cybercrime different from traditional crime? Explain the liability of Network Service Providers (Intermediaries) under the IT Act.

Answer:

Difference Between Cybercrime and Traditional Crime: Cybercrime is defined as crimes committed on the internet using the computer as a tool to target the victim. They differ from traditional crimes in several critical ways:

1. **Detection and Prosecution:** Cybercrimes are often much harder to detect, investigate, and prosecute compared to traditional crimes, causing greater overall damage to society.
2. **Physical Evidence:** In traditional crimes, criminals usually leave physical proofs like fingerprints at the crime scene. In cybercrimes, criminals commit their acts through the internet, leaving very little to no physical proof.
3. **Jurisdiction and Distance:** It is difficult to determine exactly where a particular cybercrime took place because the attacker can harm the victim while sitting at a far distance, crossing international borders seamlessly.

Liability of Network Service Providers (Intermediaries): A 'network service provider' (like an ISP or cellular mobile service) acts as an intermediary providing access to a network. Under **Section 79** of the Information Technology Act, they are generally granted exemptions (safe harbor) from liability for third-party information or data hosted by them.

Grounds for Exemption: The intermediary is NOT liable if:

- Their function is strictly limited to providing access to a communication system.
- They do not initiate the transmission, select the receiver, or modify the information contained in the transmission.
- They observe "due diligence" while discharging their duties under the Act.

When They Become Liable (Exemption Revoked): The intermediary WILL be held liable if:

1. They conspired, abetted, aided, or induced the commission of the unlawful act.
2. Upon receiving actual knowledge or being notified by the government that their computer resource is being used for an unlawful act, they fail to expeditiously remove or disable access to that material.

Question 41: What is the need for Security Management? Explain the six principles of security management and list various data security measures.

Answer:

Need for Security Management: Security management means minimizing the interruption of business activities and reducing the vulnerability to various cyber attacks. It bargains with the distinctive trust aspects of information. Security applies not just to computer systems, but to all perspectives of securing data or information in whatever structure it exists, utilizing integrated appliances, software, and surveillance.

The Six Principles of Security Management:

1. **Availability:** Ensures continuous accessibility of systems, procedures, and data for authorized customers, protecting against deliberate or inadvertent attempts to deny access.
2. **Integrity:** Guarantees that data has not been modified in an unconstitutional way and that systems are free from illicit manipulation, ensuring precision and consistency.
3. **Confidentiality:** Covers the procedures and controls utilized to secure customer and organizational data against illicit access or use.
4. **Accountability:** Incorporates controls essential to follow activities back to their source, underpinning non-repudiation, deterrence, and the legitimate tolerability of records.
5. **Assurance:** Addresses the strategies used to create certainty that specialized security measures are actually working as anticipated.
6. **Privacy:** Centers on the constitutional rights of people regarding how organizations gather, process, offer, document, and erase individual data as per the law.

Various Data Security Measures: To safeguard against evolving threats, organizations must implement the following measures:

- **Access Security:** Restricting access to authorized users only to monitor who accesses particular data.
- **Data Encryption:** Using unique encryption codes so that intercepted data remains unreadable to cybercriminals.
- **Email Security:** Using strong passwords and end-to-end encryption to prevent hackers from spreading malware or phishing attacks via email.
- **Risk-Assessment Analysis:** Proactively identifying internal and external risks to the confidentiality of personal information.
- **Strong Firewall:** Protecting the system from internet traffic and controlling who gains access to the organization's network.
- **Antivirus Protection:** Using software designed to avoid, detect, and deal with malware by running background scans.
- **Regular Back-ups:** Storing data safely offline or in alternative locations to avoid total loss of data due to destruction or unauthorized access.

Question 42: Explain the UNCITRAL Model Law, 1996 and discuss the role of filtering devices and rating systems in regulating cyberspace.

Answer:

The UNCITRAL Model Law, 1996 The United Nations Commission on International Trade Law (UNCITRAL) is a Model Law on Electronic Commerce adopted by the UN Commission in 1996. Its specific goal is to eliminate all kinds of legal and technical hindrances and to boost legal certainty for the cyber domain by postulating a series of internationally acceptable rules. India's Information Technology Act, 2000 was enacted on the grounds of this model law.

It has embraced three core doctrines:

1. **Non-discrimination principle:** No document can be denied legal enforceability simply because it is in an electronic form.
2. **Technological neutrality principle:** Adapting to existing laws which are neutral in nature so they can be made applicable to future innovations.
3. **Equivalence functional principle:** It affords uniform treatment to transactions done online exactly as that of transactions done offline.

Role of Filtering Devices and Rating Systems In the self-regulatory framework of cyberspace, where centralized government control is difficult, these tools play a massive role:

- **Filtering Devices:** These are software tools used to block access to unwanted material. Examples include e-mail filters that automatically delete 'spam', and site-blocking filters that screen out specified websites or websites containing specific keywords that relate to objectionable content.

- **Rating Systems:** These systems classify the content on an Internet site. A prominent example is the RSACI (Recreational Software Advisory Council labeling scheme for the Internet), which addresses the level of violence, sex, nudity, and language on a website. It empowers parents and schools to make their own choices without inevitably blocking useful content simply because a "bad" word was detected.

Question 43: What are Domain Name Disputes (Cybersquatting) and how are Trademarks infringed through the abuse of Search Engines?

Answer:

Domain Name Disputes (Cybersquatting): A domain name is the internet/web address of a website and acts as the identity of a business, carrying its goodwill and reputation.

- **Cybersquatting:** Also known as "domain name hijacking", it is an unscrupulous practice that constitutes the act of registering a domain name with a *malafide* (bad faith) intention which is actually someone else's trademark. People register domain names of other real owners as their own and take advantage of it by selling them back to the real trade owner at an excessive price.
- **Case Law Example:** In *Yahoo! Inc. v. Akash Arora, 1999*, the defendant created and registered a deceptively similar website domain name "YahooIndia.com". The High Court of Delhi restrained the defendant from misusing the trademark.

Trademark Infringement Through Abuse of Search Engines: Search engines (like Google and Yahoo) connect users with the web, but their mechanisms can lead to trademark infringements in two primary ways:

1. **Sponsored Ads/Keyword Purchasing:** The main source of income for search engines is advertisements. Search engines allow advertisers to purchase advertising space on the page of a trademark actually searched by users. Showing sponsored ads of competitors when a specific trademark is searched is objected to by trademark owners as an unfair trade practice that creates confusion in the mind of the consumer.
2. **Abuse of Meta Tags:** Meta tags are hidden text in a webpage's code used by search engines to scan and categorize the site. Companies can use them in a deceptive manner by placing a competitor's trademark in their hidden text to divert e-consumers. For example, a rival shoe company may bury the meta tag "Bata" in its Web page to lure internet users who are searching for Bata products.

QUESTIONS LIST

1. Explain the concept and need for cyber security. What are the pros and cons (benefits and disadvantages) of digital security?
2. Discuss the various security issues and threats emerging in Cyberspace (e.g., Distributed Denial of Service (DDoS) attack, Malwares, Crypto-jacking, Cyber-physical attack, Phishing).
3. What are the different challenges to cyber security? How can technology be used to provide solutions to cyber-attacks (like unauthorized access, malwares, etc.)?
4. Write short notes on: (a) Intrusion Detection System (b) Malwares.
5. What is meant by Cryptography? Explain the five key functions of cryptography.
6. What is Encryption? Distinguish between Symmetric key and Public key encryption schemes.
7. Discuss the advantages and disadvantages of Symmetric-key cryptography and Public-key cryptography.
8. Discuss the RSA algorithm with the help of an example.
9. Explain the types of conventional ciphers. Specifically, explain Substitution cipher and Transposition cipher with examples.
10. What is steganography? Give any two examples where the steganography technique is used.
11. Explain the concept and uses of Digital Signatures and Digital Certificates.
12. "Authentication is one of the most important of all information security objectives." Critically examine the statement.
13. Discuss the core elements/governing principles of data security (CIA Triad: Confidentiality, Integrity, Availability). Explain how confidentiality can be maintained and availability can be ensured in a data management system.
14. What is the need for security management? Explain the six principles of security management.
15. Discuss in brief the various data security measures (e.g., E-mail security, Anti-virus protection, Access security).
16. Explain the following terms: (a) Data Management (b) Security audit (c) Security policy.
17. What is the need for regulating cyberspace? Explain any two strategies for regulating the Internet.
18. Explain the role and purpose of using filtering devices and rating systems in the context of the legal and self-regulatory framework of cyberspace.
19. Explain the UNCITRAL Model Law, 1996.
20. Discuss the laws in the United States and the United Kingdom for the regulation of cyberspace.
21. Give an overview of Cyberspace regulation in India.
22. How is cybercrime different from traditional crime? List the four categories of cybercrimes and list various cybercrimes under each category.
23. Discuss the acts/circumstances under which a person (wrongdoer) is liable to pay damages/compensation under Section 43 of the Information Technology Act, 2000.
24. Discuss the offences in the context of the regulation of cyber law content in India.
25. What are the liabilities of 'Network Service Providers' (intermediaries) in the context of cybercrimes? Explain the grounds for their exemption.
26. What is Cyber Forensics? Explain the need and importance of the preservation of Digital Evidence and list Cyber Forensic Investigation tools.
27. What is the meaning of Intellectual Property? Describe in brief the various forms of Intellectual Property (e.g., Copyright, Patents, Trademarks, etc.).
28. Critically examine the copyright issues in the digital medium. Briefly explain the Universal Copyright Convention.
29. Discuss the issues arising due to Linking, Inlining, and Framing. What is framing? Explain the liability for framing with the help of case laws.
30. What is a Trademark? What are the possible trademark infringements related to search engines?
31. What is meant by Domain Name disputes (Cybersquatting)?
32. Explain Patent misuse.
33. What are the different types of remedies available against the infringement of IPR (Intellectual Property Rights) / Trademarks?
- Q34:** Discuss the acts/circumstances under which a person is liable to pay damages/compensation under **Section 43 of the Information Technology Act, 2000.**
- Q35:** What is **Cyber Forensics**? Explain the need and importance of the preservation of Digital Evidence and describe Cyber Forensic Investigation tools.
- Q36:** What do you mean by Encryption? Distinguish between **Symmetric key and Public key** encryption schemes, mentioning their advantages and disadvantages.
- Q37:** Discuss the core elements of Data Security (**The CIA Triad**). How can these principles be practically implemented in a data management system?
- Q38:** What is meant by Cryptography? Describe the **RSA algorithm** with a suitable example.
- Q39:** Discuss the issues arising due to **Linking, Inlining, and Framing** in Cyberspace. Explain the liability for framing using relevant case laws.
- Q40:** How is **cybercrime different from traditional crime**? Explain the liability of **Network Service Providers (Intermediaries)** under the IT Act.
- Q41:** What is the need for Security Management? Explain the **six principles of security management** and list various data security measures.
- Q42:** Explain the **UNCITRAL Model Law, 1996** and discuss the role of **filtering devices and rating systems** in regulating cyberspace.
- Q43:** What are **Domain Name Disputes (Cybersquatting)** and how are Trademarks infringed through the abuse of **Search Engines**?